

Umair Ali

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Professional Summary

AI Engineer pursuing an MS in Data Science with experience designing and deploying intelligent AI systems using **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **Machine Learning**, **Deep Learning**, and **Multi-Agent AI**. Skilled in building scalable AI applications with **Python**, **PyTorch**, **LangChain**, **LangGraph**, **FastAPI**, **Docker**, and **vector databases**. Experienced in developing production-ready AI pipelines, model evaluation, and LLM safety solutions with a strong interest in AI research and real-world deployment.

Experience

Associate AI Engineer | Graduate Research Assistant

Jan 2025 – Present

Information Technology University (ITU), Lahore

- Designed and developed production-ready AI applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, FastAPI, and vector databases for intelligent document understanding, semantic search, and domain-specific question answering.
- Engineered scalable AI pipelines encompassing document preprocessing, semantic chunking, embedding generation, hybrid retrieval, reranking, prompt engineering, agentic workflows, and RESTful API integration to deliver reliable and efficient AI solutions.
- Conducting research on trustworthy Generative AI by developing hallucination detection frameworks using Natural Language Inference (NLI), neural-symbolic learning, uncertainty estimation, and biomedical knowledge graphs, while translating research innovations into deployable AI systems.

Projects

ClinHallu: Medical Hallucination Detection (Master's Thesis) | Python, PyTorch, PEFT, DeBERTa [GitHub](#)

- Designed **GAER++**, a parameter-efficient detector (DeBERTa-v3 + LoRA, **11.0M** trainable parameters) that compares answer and evidence representations through cross-attention, token grounding, and agreement/disagreement features.
- Built a controlled protocol measuring source-to-benchmark shift, adding MedHallu data in 1,000-example steps.
- Reached **93.75 F1** and **98.49 AUROC** on MedHallu, **+16.64 F1** over source-only training, outperforming four zero-shot LLM judges (best **89.30 F1**), with **79.7%** of the gain from the first 1,000 benchmark examples.

AAXIS AI RAG Assistant | FastAPI, LangChain, Chroma, n8n, LangSmith, Docker

[Live Demo](#) | [GitHub](#)

- Built a public company chatbot over live PDFs with **n8n**-driven ingestion, incremental indexing (only changed chunks re-embedded), and hybrid vector + BM25 retrieval with RRF.
- Added cited answers in English and Urdu, LangSmith tracing, and an evaluation suite (Hit@k, MRR, faithfulness).

AAXIS InvoiceFlow | FastAPI, OpenCV, Tesseract OCR, LLMs, QuickBooks API

[Live Demo](#) | [GitHub](#)

- Developed an invoice-processing system that reads PDFs, scans, and emails (OpenCV deskew + OCR), validates line-item math, detects duplicates and unknown vendors, and posts approved bills to QuickBooks Online.
- Added human review with on-page evidence highlights, two-person approval, and a hash-chained audit trail.

AI-Assisted Medical Coding System | FastAPI, PostgreSQL, Qdrant, RAG, LLMs, Next.js

[GitHub](#)

- Developed a multi-stage coding pipeline (clinical NLP with negation detection, hybrid RAG over **98,262** ICD-10-CM codes, schema-constrained LLM, deterministic validation) for ICD-10-CM, CPT, and HCPCS coding.
- Added calibrated confidence, coder-only approval, a 30+ rule claim scrubber, and a tamper-evident audit log.

Technical Skills

Machine Learning: Python, SQL, Scikit-learn, XGBoost, PyTorch, TensorFlow

Generative AI: LangChain, LangGraph, CrewAI, AutoGen, Prompt Engineering, RAG, AI Agents, n8n

Large Language Models: OpenAI API, Anthropic API, Gemini API, OpenRouter, Hugging Face Transformers

Computer Vision: YOLOv8, OpenCV, OCR

Vector Databases: FAISS, Pinecone, Qdrant, Chroma, pgvector

MLOps & Deployment: FastAPI, Docker, MLflow, Git, GitHub Actions, CI/CD, AWS (EC2, S3), Render

Evaluation & AI Safety: RAGAS, TruLens, LangSmith, SHAP, LLM Evaluation, Hallucination Detection

Education

MS in Data Science

Jun 2024 – Present

Information Technology University (ITU), Lahore

BS Engineering

2019 – 2023

University of Engineering and Technology (UET), Lahore

Certifications

- Deep Learning Specialization – Coursera (Andrew Ng), 2023